Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, subscription behavior, and strategic recommendations.

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Project & Dataset Snapshot

Scope

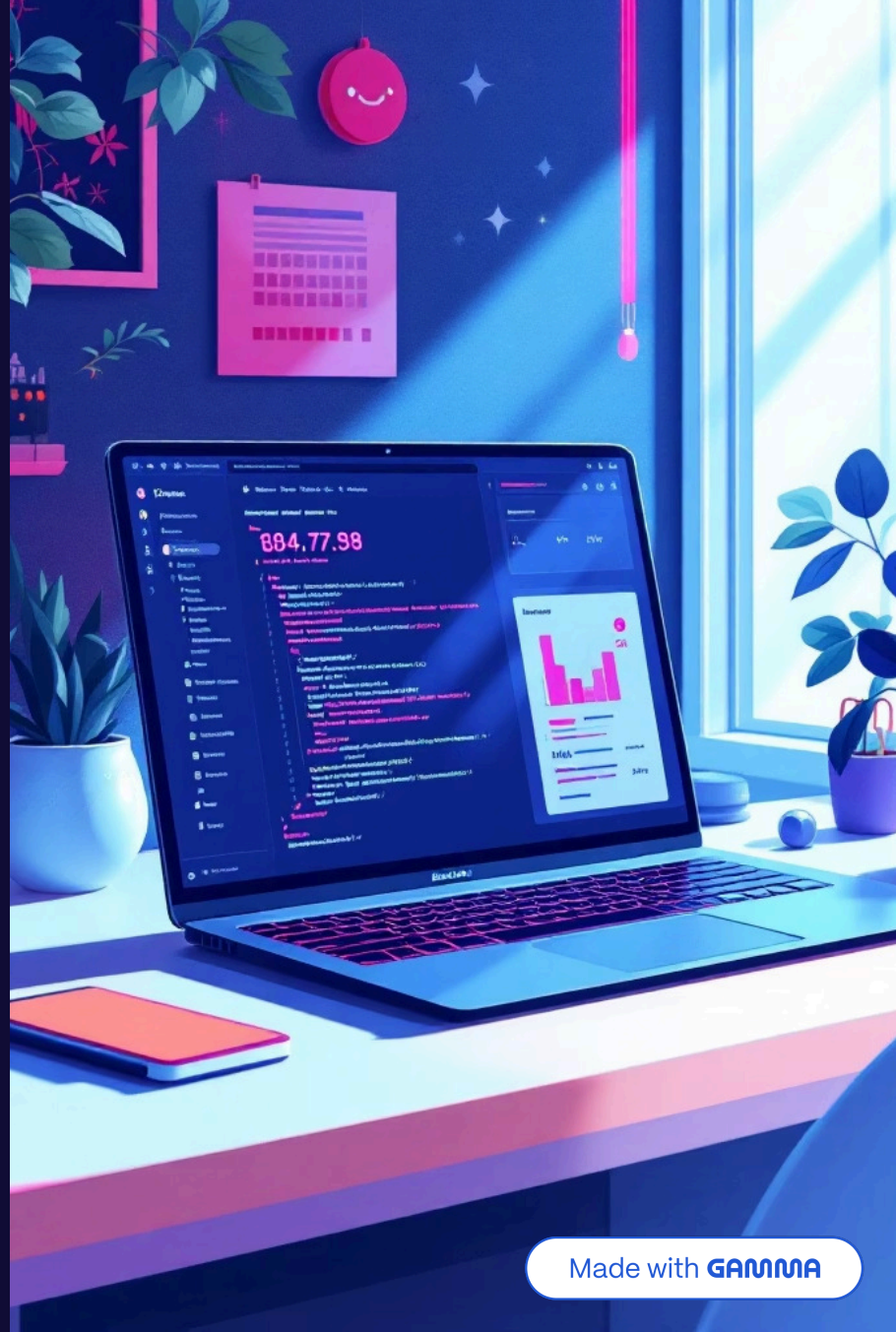
3,900 purchases across 18 columns capturing demographics, purchase details, and behavior.

Key Features

Age, Gender, Location, Item, Category, Purchase Amount, Season, Rating, Subscription, Shipping, Discounts.

Data Quality

37 missing values in Review Rating; addressed during cleaning.



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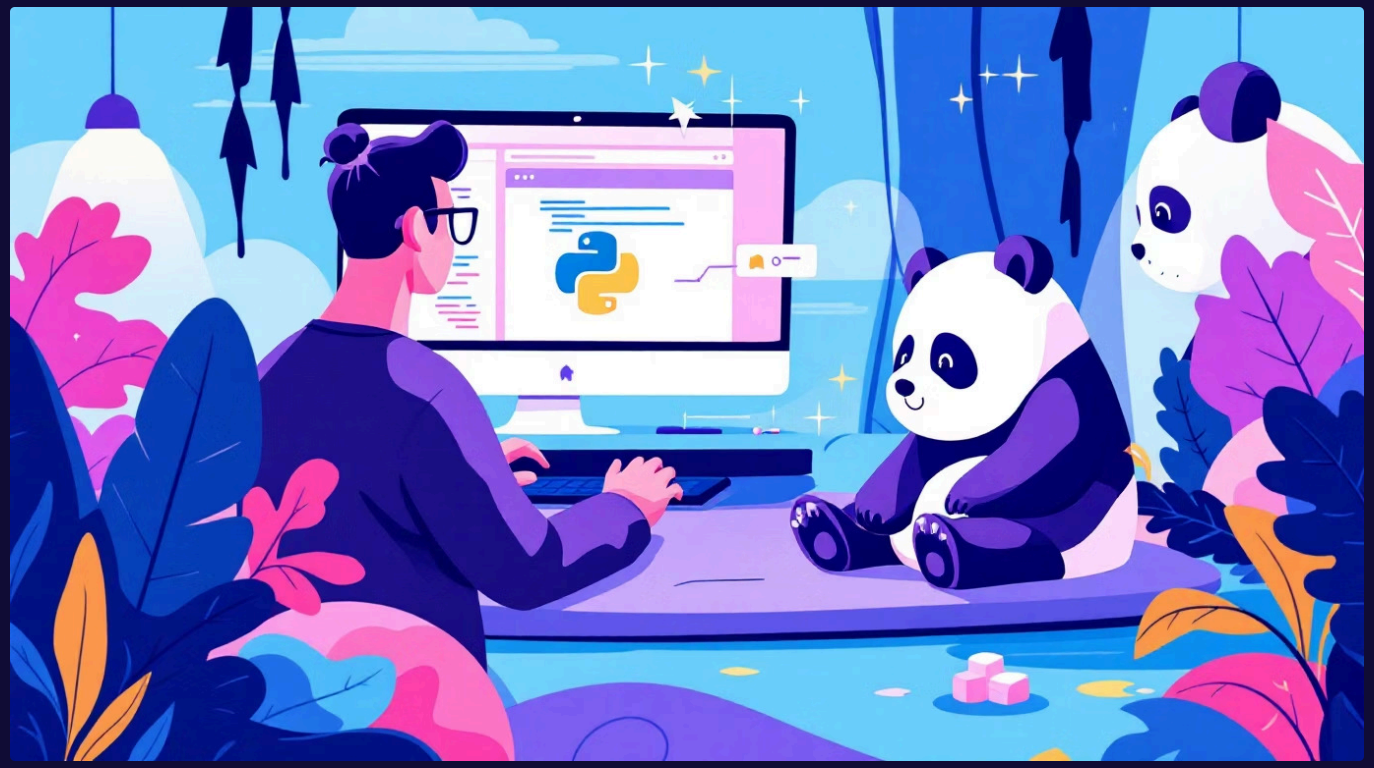
EDA — Python Workflow



- **Steps Taken**

  Data loading (pandas), df.info(), .describe().

- **Cleaning**

  Imputed Review Rating by category median; standardized column names to snake_case.

- **Feature Engineering**

  Added age_group and purchase_frequency_days; dropped redundant promo_code_used.

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Dataset Summary (Statistics)

Rows

3,900 transactions

Columns

18 features including demographics and purchase details

Avg Purchase

$59.76 average purchase amount

Avg Rating

3.75 average review rating (37 missing values)

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Business Questions — SQL Insights

Revenue by Gender

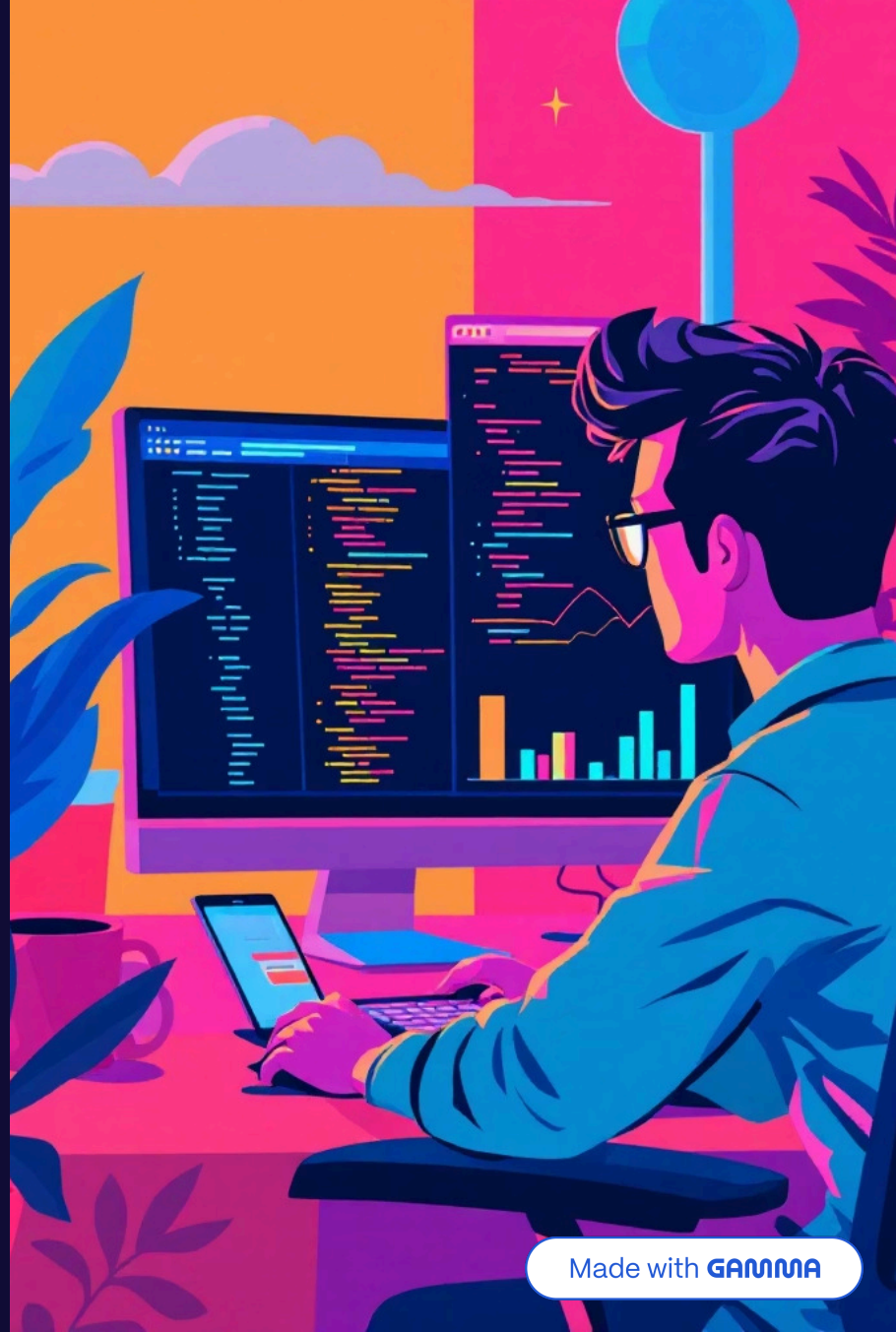
Male revenue: 157,890 • Female revenue: 75,191

Shipping Impact

Avg Standard: $58.46 • Avg Express: $60.48

Subscribers vs Non

Yes: 1,053 customers, avg $59.49 (total $62,645) • No: 2,847 customers, avg $59.87 (total $170,436)



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Product & Discount Patterns

Top Rated Items

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

Discount-Dependent

Highest discount rates: Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%

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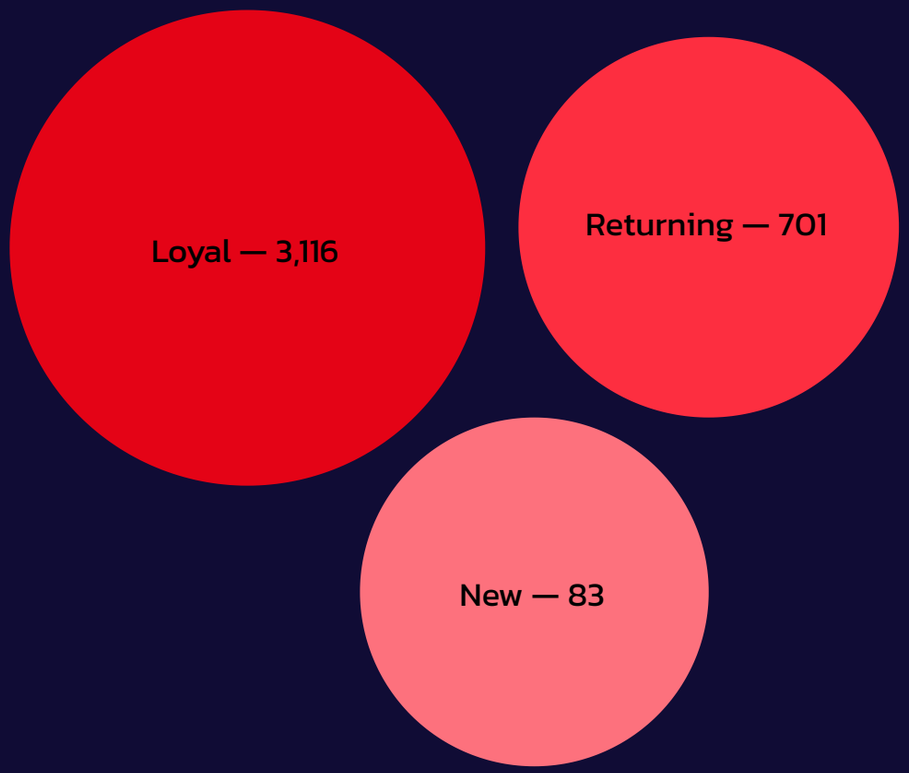
Customer Segments & Loyalty



Majority are Loyal (3,116). Returning: 701. New customers: 83. Repeat buyers (>5 purchases) lean toward subscription but most repeat buyers are non-subscribers (2,518 no vs 958 yes).

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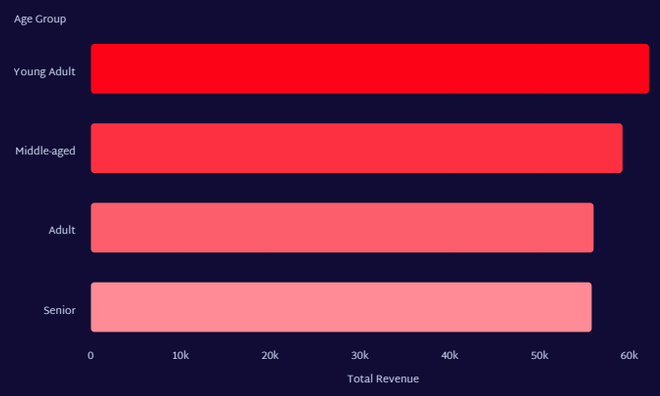
Revenue & Age Group Contribution

Young Adults contribute the most revenue (62,143), followed closely by Middle-aged and Adult groups. Use targeted campaigns to maximize ROI.



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Power BI Dashboard Overview

Interactive visuals: Subscription donut (27% Yes / 73% No), Revenue by Category (Clothing ~100K, Accessories ~70K, Footwear ~30K, Outerwear ~20K), Sales by Category and Age group comparisons for exploration.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert non-subscribers (73% of customers).

Loyalty Programs

Reward repeat buyers to deepen loyalty and lifetime value.

Review Discount Policy

Balance discount-driven volume with margin preservation for high discount items.

Targeted Marketing

Prioritize high-revenue age groups and express-shipping customers for tailored campaigns.

Product Positioning

Feature top-rated and best-selling items (e.g., Gloves, Sandals) in promotions.

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